Cluster : Security in Futuristic Technologies

Problem Statement title : Lightweight Post Quantum Messaging

Description:

Build lightweight and efficient post-quantum secure messaging systems suitable for mobile and IoT environments, ensuring confidentiality and forward secrecy in the quantum era.

Exact Deliverables :

- Prototype messaging app or protocol implementation using NIST PQC algorithms (e.g., Kyber, Dilithium).

- Lightweight key exchange protocol optimized for mobile/IoT constraints.

- Performance benchmarking reports against classical TLS messaging.

- Demonstration of secure communication across constrained devices.

Milestones, Evolution Parameters:

- Phase 1: Implement baseline PQC-based key exchange.

- Phase 2: Optimize for mobile and IoT environments.

- Phase 3: Integrate into prototype messaging app and benchmark against TLS.

- KPIs: Latency, throughput, energy usage, interoperability with existing systems.

Additional Information:

- Students should prioritize usability: PQC adoption will fail if UX is degraded.

- Explore hybrid protocols (classical + PQC) for transition scenarios.

- Align with emerging international PQC standards (NIST FIPS-203/204).